

Forced Induction and Internal Combustion Engines

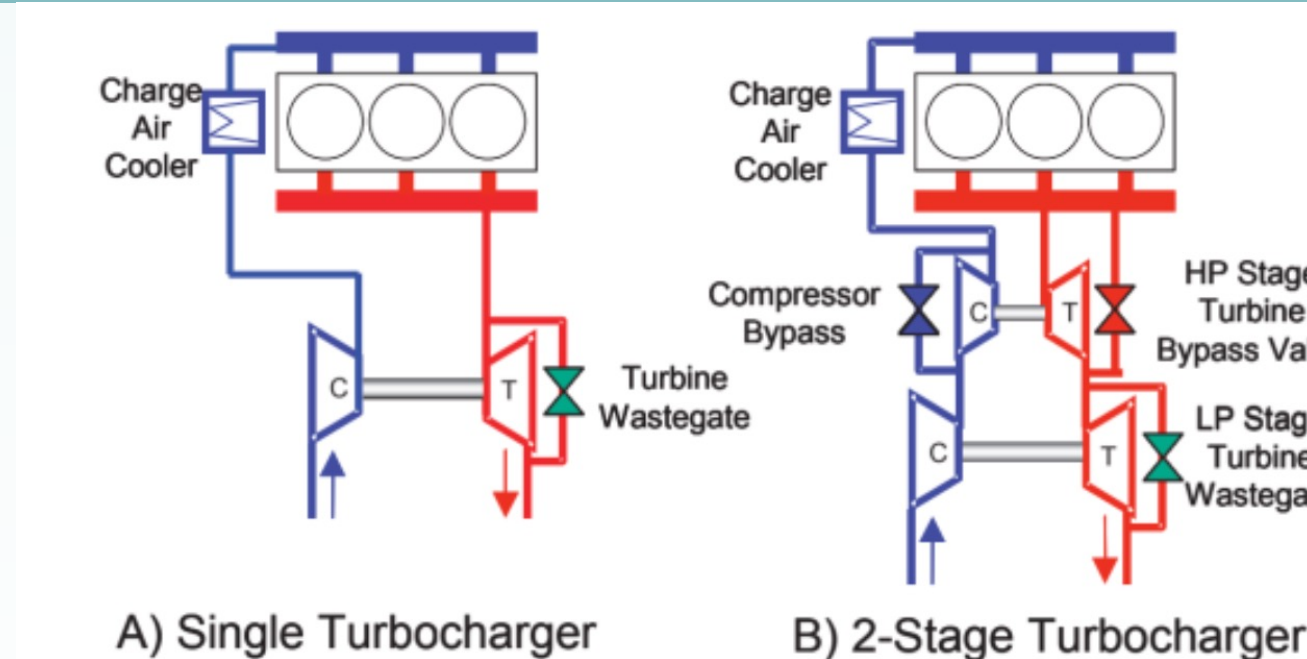


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Introduction

An internal combustion engine (ICE) runs on small, optimally timed explosions that result in a piston being forced down to drive a crankshaft. In a naturally aspirated (NA) ICE, when this piston moves downward, the vacuum created in the cylinder pulls in air at atmospheric pressure after a valve is opened by the camshaft. Fuel is injected into the combustion chamber as the air is pulled in, and a spark creates an explosion. The problem with NA is that engines need to be built to larger displacements to output comparable power to a forced induction engine, which aspirates air at increased pressures. Forced induction systems also help reduce emissions produced by the combustion process which will continue aiding manufactures in creating environmentally friendly engines.

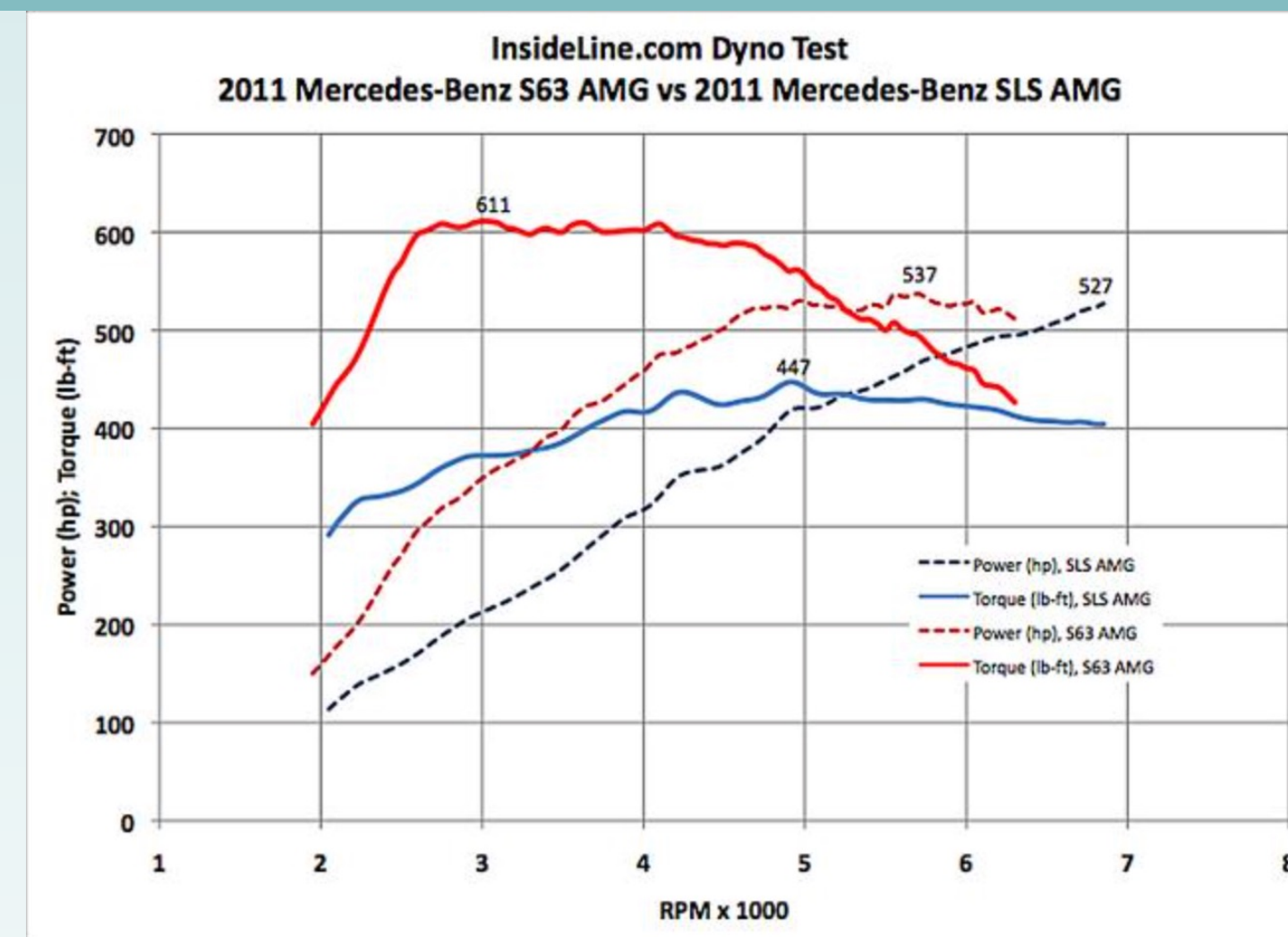
Turbocharger Systems



Credit: M. Zengeneh, Concepts in Turbocharging for Improved Efficiency and Emissions Reduction (SAE International, 2015).

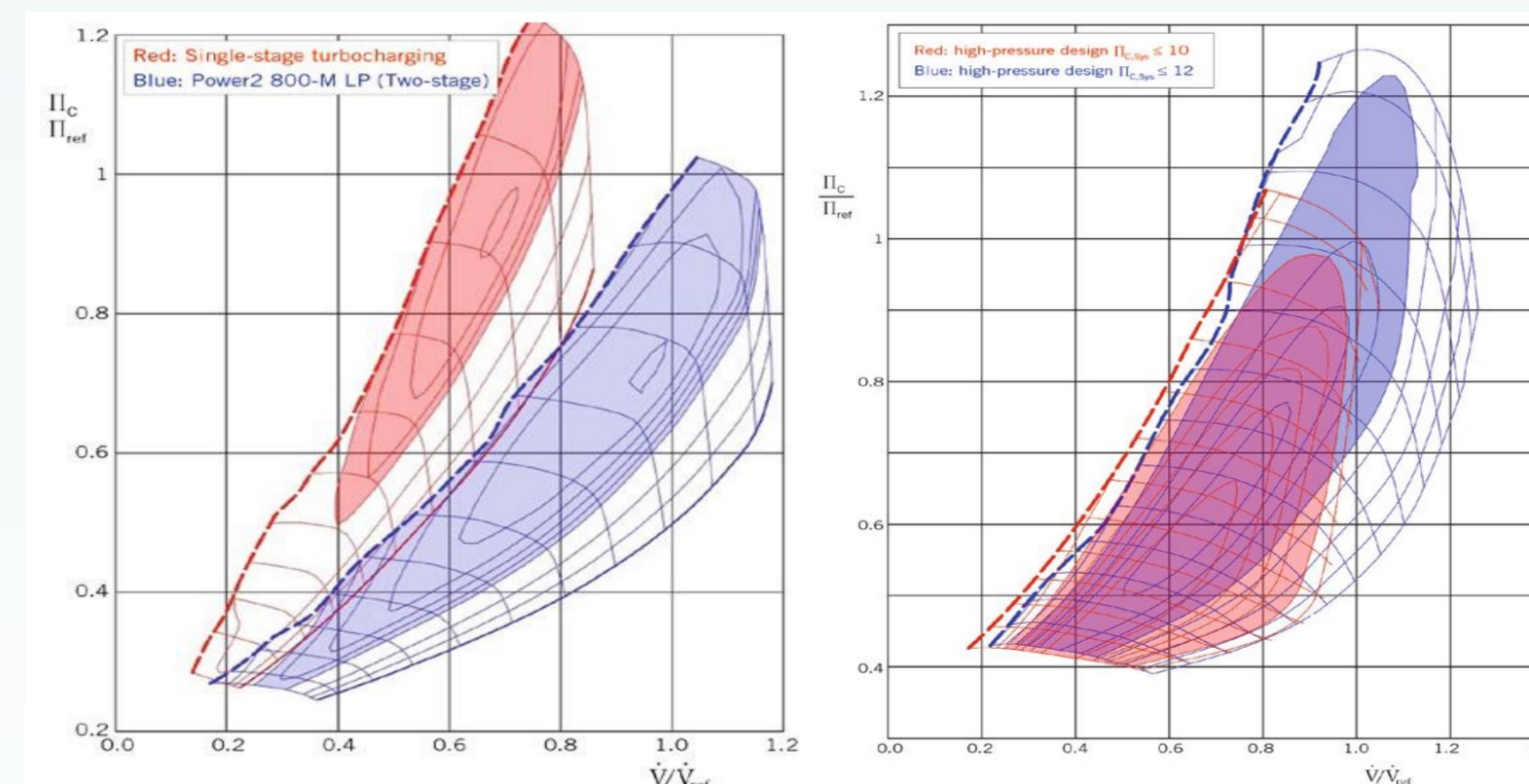
Turbochargers increase air pressure and volume into the engine by utilizing the pressure of exhaust gases from the combustion chamber of the engine to spin a set of turbines. The increase in oxygen content allows for more fuel to be injected and combusted, thus creating a more powerful combustion. This process increases fuel efficiency and allows for smaller engines which reduces emissions.

Turbochargers- Continued



Credit: Edmunds.com

- The 5.5 L twin-turbo M157 V8 is shown in red and the 6.3 L NA M159 V8 is shown in blue.
- The twin-turbo M157 has a higher power output and torque output at lower engine speeds.

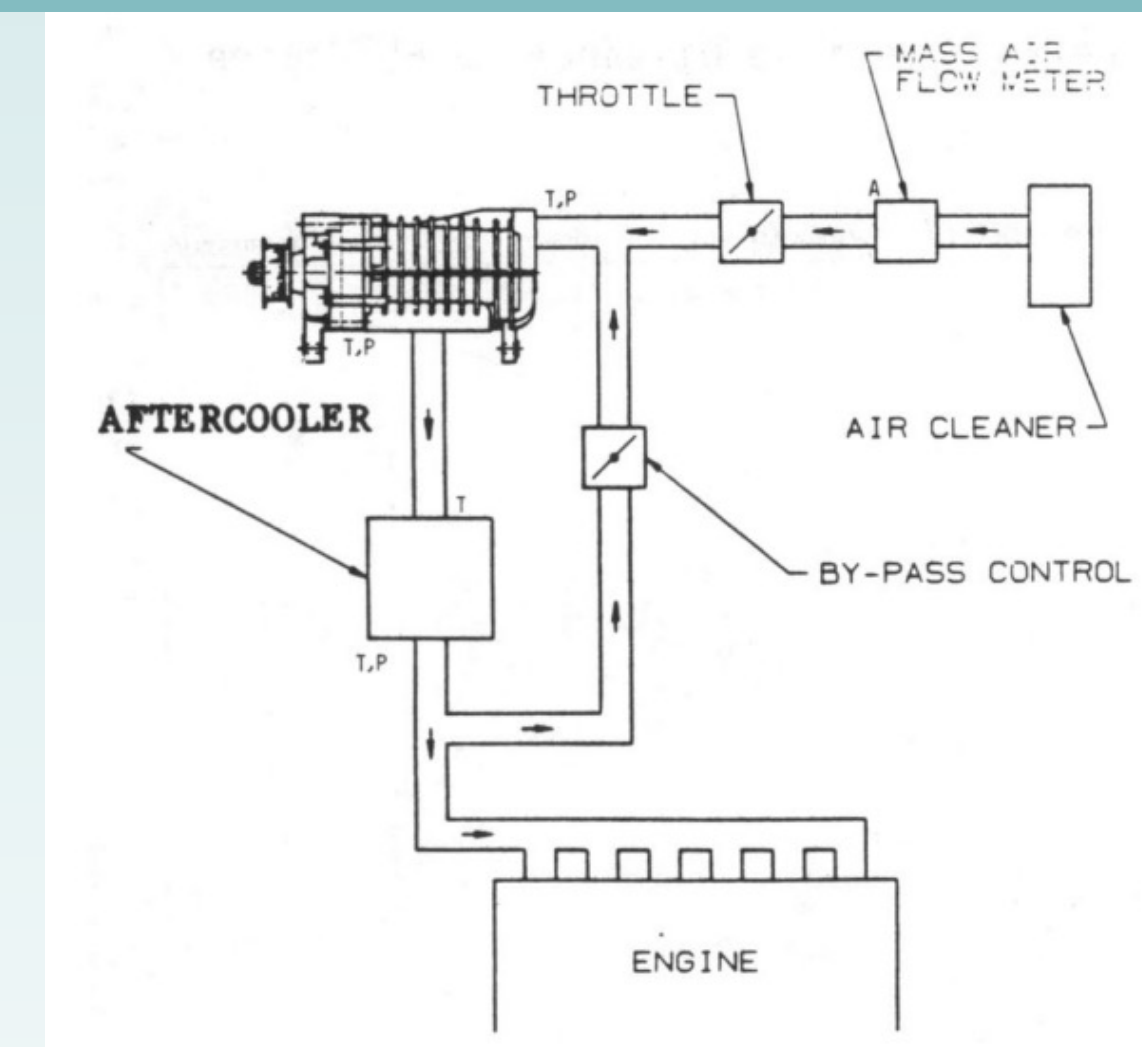


Credit: P. N. Pakale and S. U. Patel, International Journal of Research in Engineering and Technology 04, 17 (2015).

A single turbocharger takes time to provide the intake with boost pressure but can be remedied using a dual-stage setup with a second, smaller turbocharger.

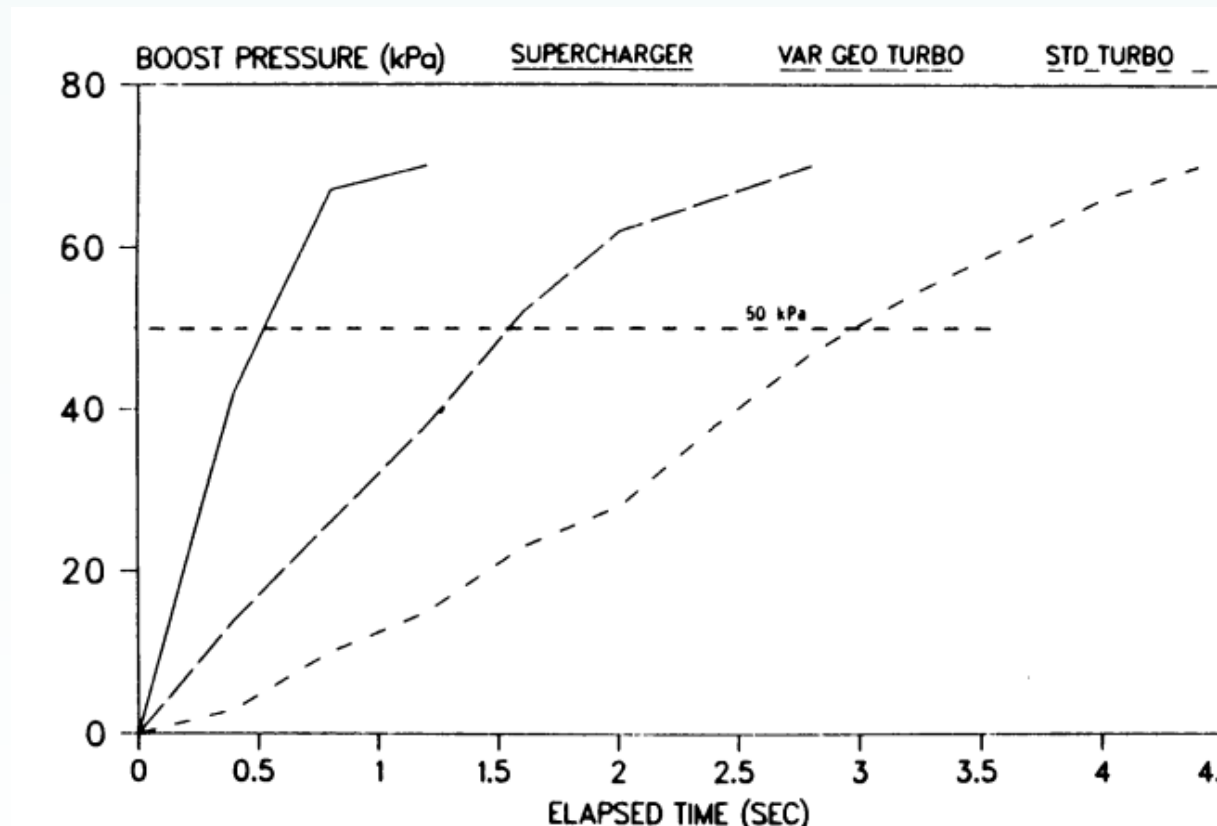
- The model on the left is for low exhaust pressures and the one on the right is for high.
- The x-axis shows the output volume of the compressor(s) while the y-axis is input pressure.

Supercharger Systems



Credit: L. H. Uthoff and J. W. Yakimow, SAE Transactions 96, 985 (1987).

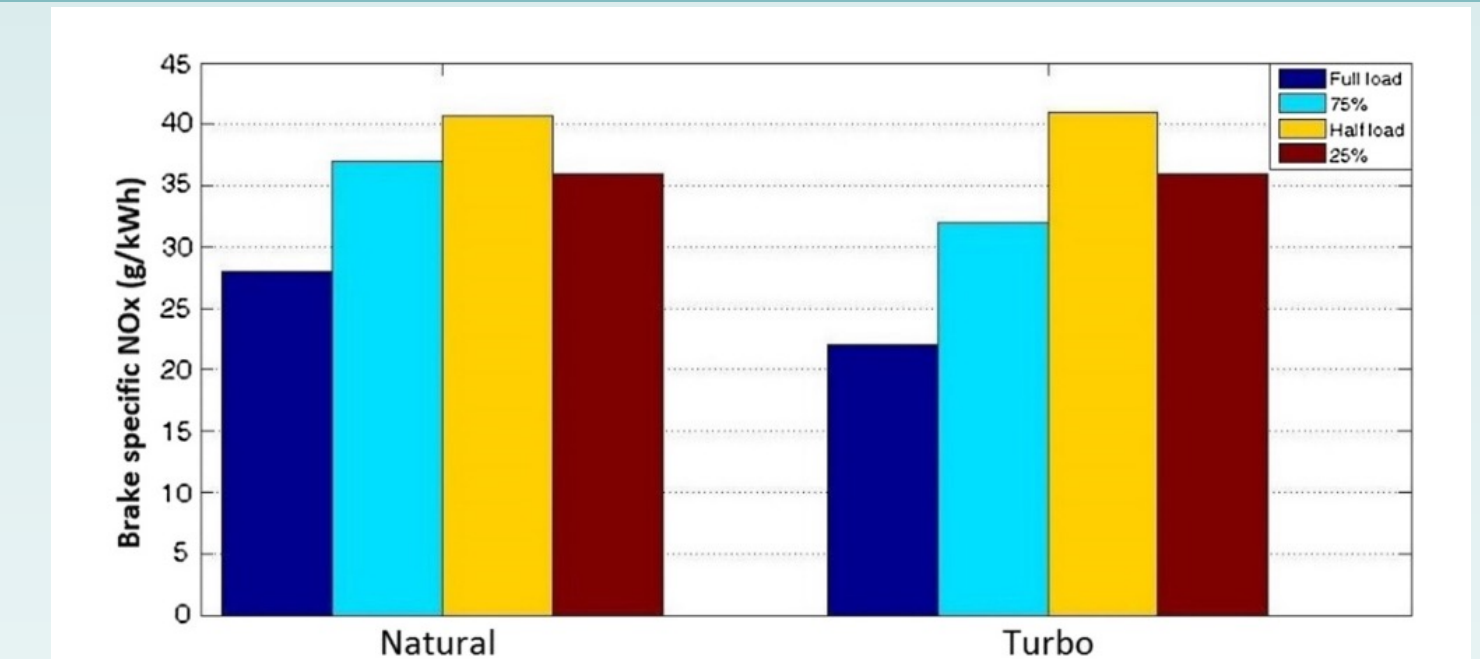
Superchargers aim to reduce the time for intake pressure to increase by getting their power to turn the turbine from the engine's crankshaft. Because the turbine is always rotating at a speed proportional to the engine speed, the intake pressure is high enough at all engine speeds to provide boost which is an improvement over turbochargers.



Credit: L. H. Uthoff and J. W. Yakimow, SAE Transactions 96, 985 (1987).

The response time for an Eaton Model 45 supercharger and two turbochargers is shown above. The supercharger provides boost pressure much faster than the turbocharger which creates faster throttle response.

Environmental Effects



Credit: A. R. Mahmoudi, I. Khazaei, and M. Ghazikhani, Alexandria Engineering Journal 56, 737 (2017).

Forced induction is not only beneficial to creating more power, but also to decrease emissions of carbon dioxide and nitrogen oxides. The figure above shows brake torque specific NO_x emissions from both a naturally aspirated and turbocharged engine with equal displacements. This figure shows the decrease in emissions from a forced induction system due to more hydrocarbons and amines in the fuel being burned from the increased oxygen content. With recent calculations on the energy requirements to transition to electric vehicles, further development of new technology such as electric turbochargers paired with hybrid electric motors could be the future for environmentally friendly automotive engineering.

Applications

The application of forced induction systems cannot be universally defined because each application can vary widely. In consumer applications, generally emissions and cost reduction are high priority, so turbocharger systems are used. Many high-performance applications implement superchargers for the instant power output, and turbochargers when peak power is of concern.